

Practice Quiz — The Innovation Ecosystem

Part A: Multiple Choice

1. In Active Inference terms, what is the primary function of a platform in an innovation ecosystem?

- A) To maximize profit for the platform owner
- B) To serve as a shared generative model that reduces free energy for multiple agents
- C) To eliminate competition through network effects
- D) To replace regulatory oversight with market mechanisms

1. Which of the following best describes the Markov blanket of an invention operating within its ecosystem?

- A) The legal boundary between the company and its competitors
- B) The separation between internal states and external states, mediated by sensory and active states
- C) The physical boundary of the product itself
- D) The set of patents that protect the invention from copying

1. During which ecosystem phase is free energy highest for new entrants?

- A) Emergence, because no dominant design has been established
- B) Growth, because competition is intensifying
- C) Maturity, because incumbents have optimized their models
- D) Decline, because the ecosystem is collapsing

1. How do regulatory frameworks function within an Active Inference model of the innovation ecosystem?

- A) As external noise that increases uncertainty for all agents
- B) As active states through which governments intervene in markets
- C) As shared priors that coordinate expectations and reduce uncertainty across the ecosystem
- D) As prediction errors that signal market failures

1. The VHS vs. Betamax outcome is best explained by Active Inference as:

- A) Betamax had a flawed generative model of video recording technology
- B) VHS achieved a lower free energy equilibrium for the ecosystem due to stronger network effects among

content producers and hardware manufacturers C) Sony's internal states were poorly optimized for competition D) The regulatory environment favored VHS over Betamax

1. What happens to an invention's Markov blanket as it scales from a startup to a global enterprise?

A) The blanket contracts as the invention becomes more focused B) The blanket remains static once product-market fit is achieved C) The blanket expands to include new sensory and active states at larger scales D) The blanket becomes irrelevant as the invention dominates its market

1. An inventor who contributes to a standards-setting process is primarily attempting to:

A) Increase the free energy of competitors B) Shape the shared generative model of the ecosystem in ways favorable to their invention C) Delay innovation to protect their current market position D) Reduce their own internal states to simplify operations

Part B: Short Answer and Analysis

1. A startup has developed a novel biodegradable packaging material. Map the three most important ecosystem components for this invention across each blanket state (internal, sensory, active, external). Explain one strategic insight that emerges from the mapping.

2. Explain why Moderna's mRNA vaccine platform could not achieve widespread adoption before the COVID-19 pandemic, using the concepts of ecosystem phase and free energy landscape. What specific change in the ecosystem's free energy landscape made rapid adoption possible?

3. You are launching a new open-source software tool that competes with established proprietary solutions. Using the Linux ecosystem as a reference, describe three ecosystem-building strategies you would pursue and explain how each reduces free energy for potential adopters and contributors.